

02_std_current_data_structure.md

Overview

Engine3 BETA is a clinical knowledge compression & navigation system organized into 10 subject folders + supporting folders. All data is **JSON or Markdown only**. Structure enforces pattern recognition via entity linking, not depth memorization.

Folder Layout (Engine3 BETA root)

```
Engine3 BETA/
├─ 01_anestesi/
├─ 02_kirurgi/
├─ 03_ortopedi/
├─ 04_endokrinkirurgi/
├─ 05_handkirurgi/
├─ 06_karlkirurgi/
├─ 07_onkologi/
├─ 08_plastikkirurgi/
├─ 09_urologi/
├─ 10_akutmed_trauma_smart/
├─ 11_exams/
├─ 12_curriculum/
├─ 13_sim_cases/          [SEPARATE FOLDER, SEE 03_std_current_work_simcases.md]
├─ .claude/
├─ .git/
├─ app.py
├─ main.py
├─ index.html
├─ script.js
├─ style.css
└─ [backend files]
```

Per-Subject Folder Structure (01-10)

Each subject folder (e.g., `02_kirurgi/`) contains exactly **4 files**:

1. `01_coverage_map_{subject}.md`

- **Purpose:** Entity index & structural overview
- **Content:**
 - List of all entities (klinik, undersökning, handläggning, differential, varning)

- Breadcrumbs/hierarchy showing how entities relate conceptually
- No questions; no detailed explanations
- Human-readable reference for what knowledge this subject covers
- **Example:** `01_coverage_map_kirurgi.md`
- **Size:** ~5-15 KB

2. `02_question_bank_{subject}.json`

- **Purpose:** Structured Q/A tied to entities
- **Content:**
 - Array of question objects
 - Each question links to one or more entities (via `entity_id`)
 - Fields: `id`, `question`, `answer_choices`, `correct`, `explanation`, `entities[]`, `yield` (HIGH/MED/LOW)
 - No narrative; pure data
- **Example:** `02_question_bank_kirurgi.json`
- **Size:** ~50-150 KB (depends on subject density)
- **Format:**

json

```
{
  "subject": "kirurgi",
  "version": "2.0",
  "questions": [
    {
      "id": "kir_001",
      "question": "63-årig man med akut buksmärt...",
      "entities": ["akut pankreatit", "amylase", "lipase"],
      "yield": "HIGH",
      "answer_choices": [...],
      "correct": 0,
      "explanation": "..."
    }
  ]
}
```

3. `03_audit_{subject}.md`

- **Purpose:** Validation & gap analysis (low priority; can be skipped)
- **Content:**
 - Coverage gaps (entities without questions)
 - Question quality notes

- Discrepancies between 01 and 02
- **Example:** `03_audit_kirurgi.md`
- **Size:** ~2-5 KB

4. `04_matrix_{subject}.md`

- **Purpose:** Entity-to-entity relations & decision space
- **Content:**
 - Defines all entities and how they connect
 - Relations: `entity_a ↔ entity_b` (differential diagnosis, test→finding, finding→action, etc.)
 - Enables traversal: given one entity, what's next?
 - Markdown table or prose format
- **Example:** `04_matrix_kirurgi.md`
- **Size:** ~15-40 KB
- **Example format:**

```
## Entitet: akut_pankreatit

**Diagnostik**:
- amylase ↑
- lipase ↑
- CT abdomen → nekrös skenhet

**Differentialer**:
- kolecystit (höger arcus)
- perforerad ulcer (fri gas)

**Handläggning**:
- NPO (näring på rör)
- aggressiva vätskor
- smärtlindring (NOT morphine)
- ERCP vid biliär origin
```

Supporting Folders

11_exams/

- Cleaned & implemented exam questions from previous courses
- Separate from simulation; reference material only
- Not part of active study pipeline (low priority)

12_curriculum/

- Official course requirements & learning outcomes
- Reference for what *should* be known
- Used to validate coverage in 01-10

13_sim_cases/ (See 03_std_current_work_simcases.md)

- Currently separate folder being developed
- Will integrate into Engine3 as folder 13 once finalized
- Contains simulation cases, engine rules, and playable interfaces

Data Flow & Linking

```
01_coverage_map_{subject}
    ↓ (defines entities)
04_matrix_{subject}
    ↓ (shows entity relations)
02_question_bank_{subject}
    ↓ (questions tied to entities)
Frontend (traversal via entity clicking, decision flow)
```

Key principle: Questions are not standalone Q/A; they are **entry points to entity networks**.

Answering a question reveals patterns across related entities.

Naming Conventions

- **Files:** `NN_descriptive_name_{subject}.format`
 - `NN` = sequence number (01-04 per subject, 11-12 supporting)
 - `descriptive_name` = `coverage_map`, `question_bank`, `audit`, `matrix`
 - `subject` = kebab-case subject name (`kirurgi`, `ortopedi`, etc.)
 - `format` = `.md` or `.json`
 - **Entity IDs** (in JSON & matrix): `lowercase_underscore_format`
 - Example: `akut_pankreatit`, `abcde_primary_survey`, `thompson_test`
 - **Question IDs:** `{subject_abbrev}_{number}`
 - Example: `kir_001`, `ort_042`, `anest_128`
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Frontend Consumption

The Python backend (`app.py`, `main.py`) and frontend (`index.html`, `script.js`) consume:

- `02_question_bank*.json` — loads Q/A directly
 - `04_matrix*.json` (future) — enables entity-linked navigation
 - Displays via `kirurgi.xyz`
 - Uses GitHub for repo, Render for backend deployment
 - Future: Firebase for user logins, leaderboards, saves
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Rules for New Content

1. **Always validate against existing structure:** If adding to subject folder, update all 4 files (`01`–`04`), not just one.
 2. **Entity consistency:** Entity ID in `02_question_bank` must exist in `04_matrix`.
 3. **No redundancy:** If information exists in `04_matrix`, don't duplicate it in `02`.
 4. **JSON-first for data, Markdown for relations:** Questions go in JSON; entity definitions & flows go in Markdown.
 5. **Versioning:** Each file should have a `version` field (e.g., `"version": "2.0"`).
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Example File Sizes & Structure

File	Typical Size	Lines	Format
<code>01_coverage_map_kirurgi.md</code>	12 KB	150–250	Markdown
<code>02_question_bank_kirurgi.json</code>	120 KB	2000–3000	JSON (minified or formatted)
<code>03_audit_kirurgi.md</code>	3 KB	50–100	Markdown
<code>04_matrix_kirurgi.md</code>	30 KB	300–500	Markdown

Current State

- **10 subjects** (01–10): Populated with coverage maps, question banks, matrices
- **11_exams:** Reference material (low priority)
- **12_curriculum:** Course requirements (reference only)
- **13_sim_cases:** In active development (see `03_std_current_work_simcases.md`)

- **Backend:** Python (Render-deployed), frontend via kirurgi.xyz
 - **Repo:** GitHub-hosted at `gigacook/gigacook.github.io`
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Instructions for AI Assistant

When working with this structure:

1. **Respect existing naming:** Use `NN_descriptive_{subject}` format; don't invent new file names
2. **Update all 4 per subject:** If modifying one subject, ensure `01`–`04` remain in sync
3. **Entity linking first:** Define entities in `04_matrix` before creating questions in `02`
4. **JSON for traversal data:** Use JSON for anything the frontend will navigate/filter
5. **Markdown for rules/narrative:** Use Markdown for documentation, rules, and conceptual flows
6. **Test in browser:** Validate that new content renders and links correctly on frontend before finalizing